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Regional bank failures and volatility transmission ☆

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ABSTRACT

We estimate the effect of the spring 2023 failures of Silicon Valley Bank and Signature Bank on the "connectedness" of US bank stock return volatilities using the forecast error variance decomposition framework of Diebold and Yilmaz (2012, 2014) and Lastrapes and Wiesen (2021). Using split-sample and time-varying VAR methods, we find that those failures significantly increased spillovers across a sample of surviving regional banks, but had only small and temporary effects on spillovers across systemically important too-big-to-fail banks. Our main findings imply that regulatory policy toward systemically important banks has been credible but that additional oversight of regional banks should be considered.

1. Introduction

On March 10, 2023 Silicon Valley Bank failed, the first closure of a federally insured bank in the US since October 2020 and the second largest in the US by nominal asset value. Signature Bank succumbed two days later, followed by the even larger failure of First Republic on May 1. The proximate cause of these failures was immediately understood to be a classic bank run on uninsured deposits, in light of rising market yields on long-term debt and concomitant declines in the value of the banks' bond and securities portfolios that were not sufficiently hedged in derivative markets. Prior to failure, all three banks reported large percentages of deposits that were uninsured — over 90% of the value of Silicon Valley Bank deposits exceeded the FDIC's $250,000 threshold — and thus susceptible to insolvency-inducing withdrawals.[1]

The purpose of this paper is to examine the effect of these failures and the ensuing panic on the "connectedness" of the US banking system — how tightly banks are integrated, and thus how susceptible they are to financial shocks in other parts of the system. We ask whether and how much volatility and uncertainty induced by these shocks altered the nature of stock return spillovers across banks in the system. The interconnectedness of banks, and the impact of bank failures on integration, is of obvious importance in understanding the extent to which financial shocks have systematic effects on risk and how policymakers should respond to financial turmoil and individual bank failures.[2]

Fig. 1 shows the recent performance of US bank stock returns relative to the overall market. The figure plots daily cumulative percent returns from October 2021 to July 2024 for the S&P 500 index (black), the S&P Regional Banks Select Industry Index (green), the average of seven large regional banks that did not fail, despite having high ratios of uninsured to insured deposits (red), and the average of eight very large and systemically important banks in the US (blue) as determined by the Federal Reserve System's Large Institution Supervision Coordinating Committee (LISCC).[3]

Prior to the spring 2023 panic, bank stocks — especially regional banks — generally followed the overall market closely according to the

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[1] Jiang et al. (2024) and Metrick (2024) examine the causes of these runs and failures in more detail.

[2] We use the terms "connectedness" and "spillovers" interchangeably throughout this study when referring to the degree of integration among the banks. Depending on context, the terms can refer to the specific spillover metrics developed and used in the paper (e.g., the total spillover from all the regional banks to a systemically important LISCC bank).

[3] The regional bank index is from S&P Global (2024). Data for individual LISCC and regional banks, which are used in our main analysis, are obtained from Yahoo Finance. More details of these data are provided below.

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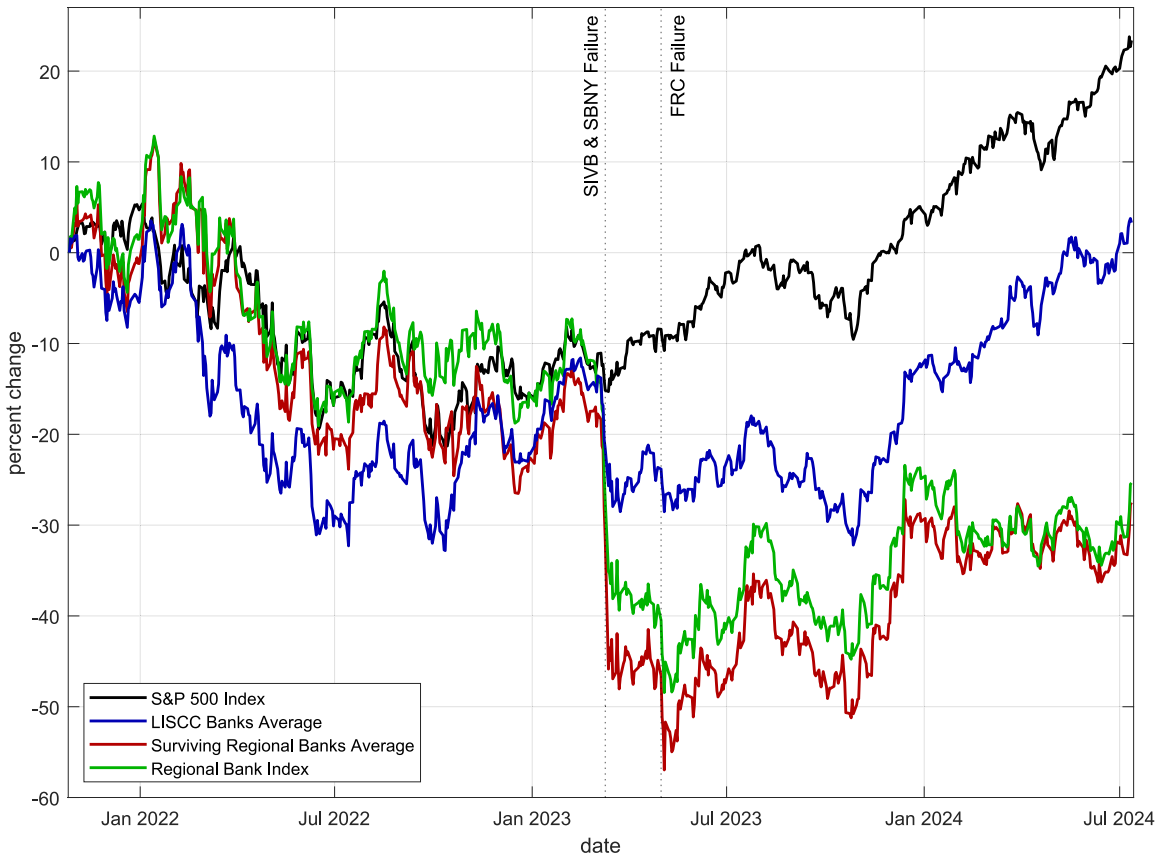


Fig. 1. Percent change in stock price since October 27, 2021 (cumulative returns). The vertical dotted lines indicate the March 10–12, 2023 failures of Silicon Valley and Signature banks and the May 1, 2023 failure of First Republic Bank. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

figure. When Silicon Valley failed, bank stock prices fell precipitously, both in absolute terms and relative to the overall market; however, those of the regional banks fell by much more. Following First Republic's failure in May, cumulative returns for the regional bank index and for our sample of regional banks with high levels of uninsured deposits were 38 and 46 percentage points below the S&P 500, respectively, while LISCC bank returns were only 18 percentage points below.[4] Since the failures, LISCC bank returns have followed a trend similar to the S&P 500 but remain below the overall market. Regional banks returns have, however, stagnated and the gap with the market has widened.

These data suggest that the regional bank failures of spring 2023 were a significant financial shock and that transmission of shocks across banks has differed for regional banking institutions relative to the largest and most systemically important ones. This difference is even more pronounced for institutions with high proportions of uninsured deposits. One of the main features of our empirical analysis is to consider the differential effects of the 2023 bank failures on these two distinct banking samples.

Our focus here is not on bank stock returns *per se*, but on the *volatility* of returns. How banks are integrated with regard to the transmission of the *magnitude* of various shocks is arguably more important for policy and regulatory purposes than the *direction* of those shocks, whether positive or negative. We estimate the extent to which shocks to daily return volatility of individual banks, measured using intra-day high, low, open, and closing transactions prices, spill over to other banks in the system. Our empirical approach is based on the connectedness metrics described in Lastrapes and Wiesen (2021), which align with the seminal work of Diebold and Yilmaz (2009, 2012, 2014, 2016).

These methods entail decomposing the forecast error variance derived from reduced-form vector autoregression (VAR) models of daily bank stock return volatilities. The extent to which the system is connected depends on the reduction of forecast error variances due to conditioning on the future paths of other bank shocks. If the banking system is only loosely connected and there are few system-wide spillovers, conditioning on other bank shocks will lead to little or no reduction in error variances. If the system is highly connected, conditioning on other bank shocks will have significant explanatory power. This approach generates quantitative estimates of connectedness and multilateral spillovers for individual banks, groups of banks, and the banking system as a whole.

Our primary aim is to understand how the regional bank failures in the spring of 2023 affected return volatility spillovers among surviving banks. We treat the failures themselves as an exogenous event and compare spillovers before and after the event estimated from VAR models that include only surviving banks. We consider two approaches. The first splits the time series sample at the event (failure) date and computes average spillovers in pre- and post-failure periods. The second follows the standard approach in the connectedness literature of estimating rolling window VAR models to compute spillover metrics that evolve over time. For each approach we include all banks — systemically important and regional — in a single VAR model; however, we treat these groups separately to determine the extent to which the bank failures had differential effects on connectedness among these two groups, as suggested by the discussion above.

The economic literature on integration, connectedness, and systemic risk in financial markets is immense and replete with numerous methods for estimating the extent to which shocks specific to a market,

[4] By comparison (and not shown in this figure), the regional bank index declined 52% from January 2, 2020 to March 23, 2020, the trough of the incipient COVID-19 pandemic, relative to a 31% decline in the S&P 500, a difference of only 21 percentage points.

institution, or sector spill over to others, or for measuring interconnectedness that arises from individual responses to common factors. Work in this area generally focuses on the transmission of shocks across markets and the nature of potential contagion that propagates systemic risk, with an eye to policies that can help mitigate such risk through monitoring and supervision. Seminal studies include Engle et al. (1990), King and Wadhwani (1990), and Bekaert and Harvey (1995). Other well-cited studies that illustrate the variety of methods and applications include Bekaert et al. (2005), Pukthuanthong and Roll (2009), Billio et al. (2012), Drehmann and Tarashev (2013), Bekaert et al. (2014), Elliott et al. (2014), Acharya et al. (2016), Adrian and Brunnermeier (2016), and Brownlees and Engle (2016). Glasserman and Young (2016) provide a good review of this vast literature, and Billio et al. (2017) compare many of the different empirical methods.

The VAR-based variance decomposition approach of Diebold and Yilmaz (2012, 2014) that we exploit and build on is perhaps the most common.[5] Their methods provide precise summary measures of how variables in an economic system are interconnected and have the advantage of characterizing the data without strong theoretical restrictions. Diebold and Yilmaz (2023) reflect on the notable popularity of the method and prospects for future work. Other relevant papers that use the technique for financial market applications include Claeys and Vašíček (2014), Apostolakis and Papadopoulos (2015), Collet and Ielpo (2018), Bostanci and Yilmaz (2020), and Bratis et al. (2020). Tabak et al. (2022, Table 1) provide a comprehensive review and summary of this literature.

An alternative to our reduced-form methods is to impose theoretical restrictions on the joint process of stock return volatilities. While a structural VAR approach makes causal inference possible, reduced-form statistical relationships are most relevant for the purpose of our study and almost all of the extant literature on econometric connectedness and its applications. Furthermore, plausible structural restrictions on the joint process of stock return volatilities are hard to come by and therefore not useful for interpreting the data. As discussed in Wiesen et al. (2018), when structural identifying assumptions are controversial or nonexistent, reduced-form methods such as the generalized forecast error variance decomposition of Koop et al. (1996) and Pesaran and Shin (1998), the generalized spillover metrics of Diebold and Yilmaz (2012, 2014), and the joint spillover metrics of Lastrapes and Wiesen (2021) are preferable because the findings from such models are compatible with *any* structural process driving the data. As stated in Diebold and Yilmaz (2014, page 121) "identifying assumptions are just that — assumptions — and any set of identifying assumptions may fail."

Our paper's main contribution is to the subset of this literature that estimates spillovers and connectedness in the banking sector, such as Tabak et al. (2022), Foglia et al. (2022), Moratis and Sakellaris (2021), Arreola Hernandez et al. (2020), McNelis and Yetman (2020), Hale and Lopez (2019), Demirer et al. (2018), and Alter and Beyer (2014). We know from Demirer et al. (2018) that banks are highly connected, both within and across countries, and that connectedness among banks tends to increase during financial crises. Tabak et al. (2022) show that spillovers across national banking sectors increased significantly with the COVID-19 pandemic, a result we confirm below for spillovers across US banks. Most of the studies on banking interconnections, such as Hale and Lopez (2019), McNelis and Yetman (2020), Egger et al. (2023), and Lai and McNelis (2024), focus on estimating connectedness among the largest US banks or "global systemically important banks," but ignore smaller, regional banks. While most studies examine bank stock return volatility as we do, others use credit default spreads (Moratis and Sakellaris, 2021; Alter and Beyer, 2014) to gauge risk.

We have found only two published papers to date that have looked at the effects of the 2023 regional bank failures with regard to connectedness in financial markets. Banerjee et al. (2024) compare Diebold–Yilmaz return and volatility spillover metrics for traditional and fintech exchange traded funds (but not for individual banks, per se) before and after the failures and generally find an increase in spillovers post-failure. Akhtaruzzaman et al. (2023) examine the contagion effects of the regional bank failures on banks and other economic groups. The paper finds that dynamic conditional correlations across bank returns, aggregated by country, increased significantly post-failure, indicating that financial contagion spread beyond the US; however, this increase in connectedness was mostly isolated to the banking sector and not other markets and types of firms.

Our paper contributes to this literature in at least three ways. Most importantly, we focus on connectedness within the US banking system, distinguishing between systemically important banks and regional banks in the US. Our finding that the Silicon Valley and Signature bank failures had substantially different effects across these two groups is novel; no other studies in this literature have considered this distinction in any context. Second, unlike the above two papers that include return volatility data only up to April or May 2023 — just a month or two after the failures — we extend the time sample to July 2024 — more than a year beyond the events — to better gauge post-failure average responses and dynamics of system-wide spillovers. Finally, we use the joint spillover metrics of Lastrapes and Wiesen (2021), which builds and improves on the original work of Diebold and Yilmaz (2012, 2014). We also use these joint metrics in a new way, computing joint spillover indices across the two different banking groups in our sample.

We find that in the immediate aftermath of the Silicon Valley and Signature bank failures, average volatility spillovers from LISCC banks to other LISCC banks only marginally and temporarily changed, whereas volatility spillovers from LISCC banks to regional banks fell considerably. More importantly, in the immediate aftermath of the failures, total volatility spillovers from all other banks to LISCC banks increased only moderately, whereas total volatility spillovers to regional banks increased substantially and was driven by shocks from other regional banks. Over the long run, volatility transmissions from LISCC to LISCC banks, from LISCC to regional banks, and from regional to LISCC banks decreased. In contrast, volatility transmissions within the group of regional banks increased.

The differential effects between the regional and LISCC banks — and the fact that LISCC banks were notably less affected — are relevant to the regulatory supervision and monitoring of LISCC banks. These "too-big-to-fail" banks have additional regulatory assessments (e.g., capital, liquidity, and risk management programs) meant to increase their resilience, which the regional banks do not have. Since the systemically important LISCC banks were considerably less affected by the 2023 bank failures than the surviving regional banks, the claim that regulatory policies toward LISCC banks to enhance their stability during periods of uncertainty is, at least in this instance, credible.

We lay out details of the empirical framework below in Section 2. The data are described in Section 2.1, including the individual banks investigated and the measure of daily volatility we use. Section 2.2 describes the spillover metrics we employ and the details of the VAR models. Section 3 presents the main empirical results of our analysis. Section 3.1 considers spillover metrics averaged over the pre- and post-failure subperiods, while Section 3.2 describes our findings for the effects of the regional bank failures on connectedness using the rolling window VAR approach. Section 3.3 checks the robustness of our main findings to some of the important modeling assumptions we make. Section 4 concludes and briefly discusses policy implications.

[5] These two papers have been cited over 8000 times in Google Scholar as of October 2024.

2. Data and spillover measures

2.1. Bank sample and data

Our baseline sample includes fifteen publicly-traded US commercial banks listed in Table 1 and separated into two groups defined by size and systemic importance. The group in the top panel includes the largest and most systemically important banks in the US as determined by the Federal Reserve's Large Institution Supervision Coordinating Committee (LISCC) as of March 2023.[6] The group in the middle panel includes seven US regional banks as categorized by Yahoo Finance's industry classification. The criteria for selecting regional banks for our sample are that, as of December 31, 2022, the bank held at least $50 billion in assets, reported at least 50% of their deposits as uninsured, and survived as a viable financial institution beyond the spring of 2023.[7] Our main results are based on VAR models that treat these fifteen banks as a unified system. However, as explained below, our methods allow for distinct transmission of shocks across the two groups in light of their apparent difference in behavior as illustrated in Fig. 1. The third panel includes summary statistics for the three banks that failed in the spring of 2023. In Section 3.3, we examine whether our decision to exclude failed banks' data from the baseline sample affects our results.

The variable of interest in our analysis is daily stock return volatility. To measure daily volatility, we rely on the efficient and robust estimator developed by Yang and Zhang (2000), which exploits daily open, high, low, and closing transactions prices. In particular, for each bank in the sample we construct the log standard deviation of daily returns in annualized percent, $ln(\sigma_t)$, according to[8]

$$\sigma_t = 100\sqrt{V_{o,t}^2 + \theta V_{c,t}^2 + (1-\theta)V_{rs,t}^2} \tag{1}$$

where

$$V_{o,t}^2 = \frac{N}{n-1}\sum_{i=0}^{n-1}\left[ln\left(\frac{O_{t-i}}{C_{t-i-1}}\right) - \frac{1}{n}\sum_{i=0}^{n-1} ln\left(\frac{O_{t-i}}{C_{t-i-1}}\right)\right]^2, \tag{2}$$

$$V_{c,t}^2 = \frac{N}{n-1}\sum_{i=0}^{n-1}\left[ln\left(\frac{C_{t-i}}{O_{t-i}}\right) - \frac{1}{n}\sum_{i=0}^{n-1} ln\left(\frac{C_{t-i}}{O_{t-i}}\right)\right]^2, \tag{3}$$

$$V_{rs,t}^2 = \frac{N}{n}\sum_{i=0}^{n-1}\left[ln\left(\frac{H_{t-i}}{O_{t-i}}\right) ln\left(\frac{H_{t-i}}{C_{t-i}}\right) + ln\left(\frac{L_{t-i}}{O_{t-i}}\right) ln\left(\frac{L_{t-i}}{C_{t-i}}\right)\right]. \tag{4}$$

Here, O_t, H_t, L_t, and C_t are, respectively, the open, high, low, and closing transactions prices for a sample bank's stock price on day t. We set $n = 2$ (the shortest window available), $N = 250$ days to annualize the measure, and $\theta = 0.0783$, its default value for $n = 2$. This measure of daily volatility is preferable to alternatives — like the standard deviation of all intraday log price increments based on transactions data — because it can efficiently and consistently account for potentially large jumps in returns during non-trading hours, from the previous day's close to the current day's open.[9]

We obtain daily open, high, low, and closing stock prices for the banks in our sample from Yahoo Finance over the period January 2, 2019 to July 12, 2024, which spans the 2023 failures of Silicon Valley, Signature, and First Republic. Table 1 reports summary statistics for return volatility over the full sample, as well as the percentage of deposits uninsured at the end of December 2022. Among regional banks, the three failed banks had the highest percentage of uninsured deposits. The table indicates that, on average, daily return volatility of the regional banks has been larger than for the LISCC banks; the difference in $\bar{\sigma}$ across the average of the two groups is about nine annualized percentage points.

Fig. 2 plots $ln(\sigma_t)$ for each bank according to group. The large spikes in daily volatility around the failure dates are evident. The figure also shows that the surviving regional banks' responses to the failures exceed the reactions to the 2020 COVID-19 pandemic. More importantly, while the regional and LISCC banks responded quite uniformly to the COVID-19 pandemic, the two groups responded very differently to the 2023 bank failures.

2.2. Spillover metrics

To measure the transmission of volatility across banks, we rely on the joint spillover metrics of Lastrapes and Wiesen (2021) and the joint forecast error variance decomposition of Wiesen and Beaumont (2024), which build on the seminal work of Diebold and Yilmaz (2012, 2014). Let $\boldsymbol{Y}_t$ denote the vector stochastic process that includes individual bank equity return volatilities, $ln(\sigma_{i,t})$, for $i = 1, \ldots, K$ and t denotes days. Our baseline sample includes eight LISCC banks and seven surviving regional banks, so $K = 15$. We represent this system as a stationary reduced-form vector moving average process:

$$\boldsymbol{Y}_t = \boldsymbol{\mu} + \sum_{h=0}^{\infty} \boldsymbol{A}_h \boldsymbol{\epsilon}_{t-h}, \tag{5}$$

where $\boldsymbol{\epsilon}_t$ is a mean-zero white noise vector of shocks with covariance matrix $\boldsymbol{\Sigma}_\epsilon = E(\boldsymbol{\epsilon}_t\boldsymbol{\epsilon}_t')$, and $\boldsymbol{A}_h$ for all h are the vector moving average coefficient matrices with $\boldsymbol{A}_0$ being the $K \times K$ identity matrix.[10] The H-step ahead reduced-form forecast errors implied by the model are

$$\begin{aligned}\boldsymbol{\xi}_t(H) &\equiv \boldsymbol{Y}_{t+H} - E(\boldsymbol{Y}_{t+H}|\boldsymbol{Y}_t, \boldsymbol{Y}_{t-1}, \boldsymbol{Y}_{t-2}, \ldots) \\ &= \sum_{h=0}^{H-1} \boldsymbol{A}_h \boldsymbol{\epsilon}_{t+H-h},\end{aligned} \tag{6}$$

with corresponding forecast error covariance matrix

$$E\left(\boldsymbol{\xi}_t(H)\boldsymbol{\xi}_t'(H)\right) = \sum_{h=0}^{H-1} \boldsymbol{A}_h \boldsymbol{\Sigma}_\epsilon \boldsymbol{A}_h'. \tag{7}$$

The generalized forecast error variance decomposition, developed by Koop et al. (1996) and Pesaran and Shin (1998) and exploited by Diebold and Yilmaz (2012, 2014), accounts for the relative reduction in the H-step forecast error variance of variable i after conditioning on the future reduced-form shocks of variable j:

$$\begin{aligned}\zeta_{ij}^{gen}(H) &= \frac{E\left(\xi_{i,t}^2(H)\right) - E\left[\xi_{i,t}(H) - E\left(\xi_{i,t}(H)|\epsilon_{j,t+1}, \ldots, \epsilon_{j,t+H}\right)\right]^2}{E\left(\xi_{i,t}^2(H)\right)} \\ &= \frac{\sum_{h=0}^{H-1}(\boldsymbol{e}_i'\boldsymbol{A}_h\boldsymbol{\Sigma}_\epsilon\boldsymbol{e}_j)^2}{(\boldsymbol{e}_j'\boldsymbol{\Sigma}_\epsilon\boldsymbol{e}_j)\sum_{h=0}^{H-1}\left(\boldsymbol{e}_i'\boldsymbol{A}_h\boldsymbol{\Sigma}_\epsilon\boldsymbol{A}_h'\boldsymbol{e}_i\right)}\end{aligned} \tag{8}$$

where $\boldsymbol{e}_i$ is a K-dimensional column vector of all zeros except for a 1 in the ith row. $\zeta_{ij}^{gen}(H)$ can be interpreted as the proportion of variable

[6] Board of Governors (2023).

[7] Uninsured deposit data are from Hayes (2023).

[8] Taking the natural log of σ_t helps ensure stability of the VAR models and, as discussed in Andersen et al. (2003), brings volatility closer to a Gaussian distribution.

[9] Compared to earlier estimators also based upon open, high, low, and closing prices (e.g., Rogers and Satchell (1991), Garman and Klass (1980), and Parkinson (1980)), the volatility estimator of Yang and Zhang (2000) offers some advantages. Namely, it is a minimum-variance unbiased estimator independent of both drift and opening jumps. To compute the volatility estimators, we utilize the `volatility` function from the TTR R package by Joshua Ulrich (Ulrich, 2021). For markets that trade continuously, like cryptocurrency markets, daily measures that use all transactions price data during the day, rather than open, high, low, and closing prices might be ideal.

[10] It is possible to interpret the reduced-form shocks ϵ_t as linear combinations of underlying structural shocks, such as $\boldsymbol{\epsilon}_t = \boldsymbol{D}\boldsymbol{u}_t$, where the structural shocks $\boldsymbol{u}_t$ are orthogonal and $\boldsymbol{D}$ is a $K \times K$ matrix. As we note in the introduction, the reduced-form approach has significant advantages for our purposes.

Table 1
Summary statistics for bank return volatility.

LISCC Banks	Uninsured	$\overline{\sigma}_t$	$sd(\sigma_t)$	$\overline{ln\sigma}_t$	$sd(ln\sigma_t)$
Bank of America (BAC)	46.1%	29.292	21.330	3.246	0.469
Bank of NY Mellon (BK)	92.0%	27.079	18.983	3.167	0.474
Citigroup (C)	73.3%	30.606	24.027	3.280	0.478
Goldman Sachs (GS)	47.6%	27.139	18.195	3.186	0.440
JP Morgan Chase (JPM)	52.0%	25.311	20.101	3.084	0.491
Morgan Stanley (MS)	<45.0%	29.116	21.300	3.240	0.470
State Street (STT)	91.2%	32.264	23.205	3.339	0.476
Wells Fargo (WFC)	<45.0%	30.392	21.822	3.274	0.490
Surviving Regional Banks	Uninsured	$\overline{\sigma}_t$	$sd(\sigma_t)$	$\overline{ln\sigma}_t$	$sd(ln\sigma_t)$
Comerica (CMA)	60.4%	40.747	36.633	3.547	0.507
Cullen/Frost (CFR)	53.6%	31.656	23.315	3.314	0.481
KeyBank (KEY)	50.0%	37.870	31.847	3.479	0.502
Synovus (SNV)	50.7%	38.234	28.891	3.496	0.493
US Bank (USB)	51.4%	29.299	22.283	3.231	0.496
Western Alliance (WAL)	56.3%	49.198	89.536	3.656	0.553
Zions (ZION)	52.2%	38.812	35.170	3.512	0.487
Failed Regional Banks	Uninsured	$\overline{\sigma}_t$	$sd(\sigma_t)$	$\overline{ln\sigma}_t$	$sd(ln\sigma_t)$
Silicon Valley Bank (SIVB)	93.8%	44.933	35.048	3.658	0.499
Signature Bank (SBNY)	89.3%	43.195	27.244	3.625	0.515
First Republic (FRC)	67.4%	39.384	88.615	3.321	0.625

Percentage of uninsured deposits, sample means and standard deviations of daily return volatility (σ_t in annualized percent), and the natural log of daily return volatility. The data range for return volatilities of all LISCC and surviving regional banks is 1/2/2019 to 7/12/2024. The range for failed banks has the same start date but ends as the banks failed and trading halted: 3/10/2023 for SIVB and SBNY, and 4/28/2023 for FRC.

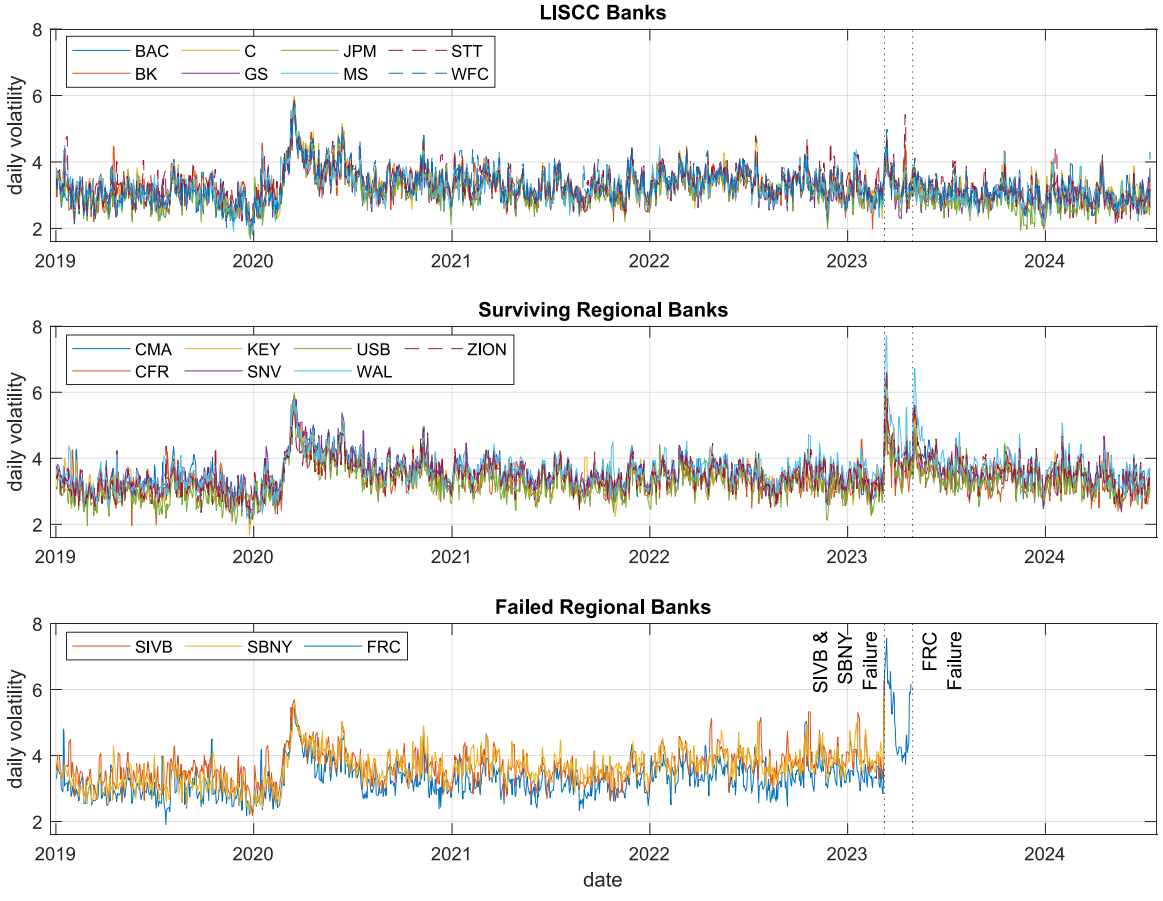


Fig. 2. Equity market volatility $ln(\sigma_t)$ for the 8 LISCC banks (top panel), the 7 surviving regional banks in the sample (middle panel), and the 3 failed regional banks (bottom panel). The vertical dotted black lines indicate the March 10–12, 2023 failures of Silicon Valley Bank and Signature Bank and the May 1, 2023 failure of First Republic Bank.

i's H-step forecast error variance that can be explained by the future shocks of variable j alone. Computing Eq. (8) for all i and j yields the $K \times K$ generalized forecast error variance decomposition matrix, $\boldsymbol{\zeta}^{gen}(H)$.

Because this generalized forecast error variance decomposition does not rely on orthogonalized shocks, the sum of all K individually decomposed components for any variable i is not restricted to sum to one; that is, $\sum_{j=1}^{K} \zeta_{ij}^{gen}(H) \neq 1$.[11] To make the variance decomposition reflect relative shares, rather than just proportionate reductions in variance, Diebold and Yilmaz (2012, 2014) normalize each row of the decomposition in Eq. (8) to obtain the elements of their "generalized spillover table" whose (i, j) element is:

$$gSOT_{ij} = \frac{\zeta_{ij}^{gen}(H)}{\sum_{\ell=1}^{K} \zeta_{i\ell}^{gen}(H)}, \quad i, j = 1, \dots, K. \tag{9}$$

Each row of the $\boldsymbol{gSOT}$ matrix comprising the $K \times K$ elements defined by Eq. (9) sums to one.[12] Diebold and Yilmaz (2014, equation 2) interpret their generalized total spillover from all others to variable i,

$$\mathcal{S}_{\bullet \to i}^{gen} = 1 - gSOT_{ii} = \sum_{\substack{j=1 \\ j \neq i}}^{K} gSOT_{ij}, \tag{10}$$

as measuring how "connected" variable i is to the other variables in the system at horizon H. The average of this measure across the K variables, the generalized spillover index ($gSOI$), is an index of system-wide connectedness:

$$gSOI = \frac{1}{K} \sum_{i=1}^{K} \mathcal{S}_{\bullet \to i}^{gen}. \tag{11}$$

The larger the index, the more connected are the variables within the system.

These measures of connectedness are intuitive; however, their proper interpretation depends on the arbitrary normalization in Eq. (9). If the objective is to understand how variable i is connected to all other variables in the system — how system-wide shocks spill over to variable i — it is more natural to condition on those variables *jointly*, as in Lastrapes and Wiesen (2021) and Wiesen and Beaumont (2024), rather than averaging over bilateral conditioning sets. As discussed in those papers, the joint approach requires no arbitrary normalization and has a straightforward statistical interpretation as a measure of system-wide connectedness and spillovers. See Lastrapes and Wiesen (2021) for a full analysis of the nature and advantages of the joint spillover index.

We estimate joint spillovers to variable i from some subset of variables $\mathbb{J}$ according to the measure

$$\begin{aligned} \mathcal{S}_{\mathbb{J} \to i}^{jnt} &= \frac{E\left(\xi_{i,t}^2(H)\right) - E\left[\xi_{i,t}(H) - E\left(\xi_{i,t}(H) | \boldsymbol{\epsilon}_{\mathbb{J},t+1}, \dots, \boldsymbol{\epsilon}_{\mathbb{J},t+H}\right)\right]^2}{E\left(\xi_{i,t}^2(H)\right)} \\ &= \frac{\sum_{h=0}^{H-1} \boldsymbol{e}_i' \boldsymbol{A}_h \boldsymbol{\Sigma}_\epsilon \boldsymbol{e}_{\mathbb{J}} (\boldsymbol{e}_{\mathbb{J}}' \boldsymbol{\Sigma}_\epsilon \boldsymbol{e}_{\mathbb{J}})^{-1} \boldsymbol{e}_{\mathbb{J}}' \boldsymbol{\Sigma}_\epsilon \boldsymbol{A}_h' \boldsymbol{e}_i}{\sum_{h=0}^{H-1} \boldsymbol{e}_i' \boldsymbol{A}_h \boldsymbol{\Sigma}_\epsilon \boldsymbol{A}_h' \boldsymbol{e}_i}, \end{aligned} \tag{12}$$

where the size of the subset $\mathbb{J}$ is $1 \leq \#\mathbb{J} \leq K$. In Eq. (12), $\boldsymbol{\epsilon}_{\mathbb{J},t+1}$ is the $\#\mathbb{J}$-dimensional vector of time $t+1$ shocks for the variables in set $\mathbb{J}$ and $\boldsymbol{e}_{\mathbb{J}}$ is the $K \times \#\mathbb{J}$ selection matrix comprised of the set $\mathbb{J}$ columns of the identity matrix. Importantly, while there is nothing mathematically preventing us from including "own" shocks in the conditioning set of Eq. (12), for the purposes of this paper we always exclude them. Namely, if we wish to measure the joint total spillover from set $\mathbb{J}$ to variable i, the shocks of variable i will always be excluded from the conditioning set.[13]

Choosing $\mathbb{J}$ to include *all* $K-1$ non-i shocks is the approach proposed in Lastrapes and Wiesen (2021) as an alternative to the original generalized metrics of Diebold and Yilmaz (2012, 2014); in this case, we denote $\mathcal{S}_{\mathbb{J} \to i}^{jnt} = \mathcal{S}_{\bullet \to i}^{jnt}$ to follow the notation of the former paper.[14] This statistic is bounded by zero and one, and is properly interpreted as the share of forecast error variance of variable i at horizon H explained jointly by the shocks of all other variables in the system.[15] Analogous to the original approach of Diebold and Yilmaz in Eq. (11), the joint spillover index is the average of $\mathcal{S}_{\bullet \to i}^{jnt}$ across all variables in the system:

$$jSOI_{\bullet|\bullet} = \frac{1}{K} \sum_{i=1}^{K} \mathcal{S}_{\bullet \to i}^{jnt}, \tag{13}$$

which measures the overall connectedness in the system. We can also compute averages across bank subsamples:

$$\begin{aligned} jSOI_{L|\bullet} &= \frac{1}{k_L} \sum_{i \in L} \mathcal{S}_{\bullet \to i}^{jnt} \\ jSOI_{R|\bullet} &= \frac{1}{k_R} \sum_{i \in R} \mathcal{S}_{\bullet \to i}^{jnt}, \end{aligned} \tag{14}$$

where $i \in L$ and $i \in R$ indicate that the sum is only over the LISCC banks or only over the surviving regional banks, respectively. $k_L = 8$ is the number of LISCC banks, $k_R = 7$ is the number of surviving regional banks, and $K = k_L + k_R$. Thus, $jSOI_{L|\bullet}$ (or $jSOI_{R|\bullet}$) is the mean total spillover from all other banks to the LISCC (or regional) banks.

As noted in the introduction, LISCC and regional banks responded differently to the bank failure shocks of 2023 and so may be interconnected in different ways. It is therefore of interest to examine how shocks are transmitted within and between the two categories of banks. To do so using our joint spillover metrics, we choose the conditioning set $\mathbb{J}$ for each bank i appropriately. For any bank i among the LISCC banks, we compute the joint spillover measure in Eq. (12) for set $\mathbb{J}$ only including the shocks of the seven surviving regional banks in the sample.[16] This statistic measures the share of the forecast error variance of each LISCC bank jointly explained by shocks of the regional banks. We can compare this statistic to the share of the forecast error variance of each LISCC bank jointly explained by other LISCC banks' shocks. Likewise, for each regional bank, we compute the same measures after

[11] This is easily seen for variable $i = 1$ when $K = 2$ and $H = 1$: $\sum_{j=1}^{2} \zeta_{1j}^{gen}(1) = 1 + \frac{(\boldsymbol{e}_1' \boldsymbol{\Sigma}_\epsilon \boldsymbol{e}_2)^2}{(\boldsymbol{e}_2' \boldsymbol{\Sigma}_\epsilon \boldsymbol{e}_2)(\boldsymbol{e}_1' \boldsymbol{\Sigma}_\epsilon \boldsymbol{e}_1)} \neq 1$. A sufficient condition for this value to equal one is that $\boldsymbol{\Sigma}_\epsilon$ is diagonal, a condition not likely to hold in general.

[12] The $\boldsymbol{gSOT}$ and the other spillover metrics are dependent on the forecast horizon H, but we suppress that in the notation for conciseness.

[13] For example, say $K = 5$ and we wish to quantify the joint spillovers coming from subset $\mathbb{J} = \{3, 4, 5\}$ to variable i following Eq. (12). If $i = 1$ or $i = 2$, then $\boldsymbol{e}_{\mathbb{J}} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$. If $i = 3$, then $\boldsymbol{e}_{\mathbb{J}} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$. If $i = 4$, then $\boldsymbol{e}_{\mathbb{J}} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}$. And if $i = 5$, then $\boldsymbol{e}_{\mathbb{J}} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$. Notice how column i of the identity matrix is always excluded from $\boldsymbol{e}_{\mathbb{J}}$ so that the shocks of variable i are omitted from the conditioning set.

[14] In this case, $\mathcal{S}_{\bullet \to i}^{jnt}$ is computed using Eq. (12) where $\boldsymbol{e}_{\mathbb{J}}$ is a $K \times K-1$ matrix equal to the identity matrix with column i removed.

[15] Note that $\mathcal{S}_{\bullet \to i}^{jnt}$ can be interpreted as the coefficient of determination (R^2) from the multiple regression of $\xi_{i,t}(H)$ on all non-i shocks jointly, whereas $\mathcal{S}_{\bullet \to i}^{gen}$ is the normalized sum of R^2s from $K-1$ "simple" regressions that condition on each non-i shock individually. For more discussion on the R^2 interpretation, see Lastrapes and Wiesen (2021, section 5.3) and Wiesen et al. (2018, section 4). The R^2 goodness-of-fit interpretation has, in turn, inspired the connectedness methods of Gabauer et al. (2023) and Balli et al. (2023).

[16] In this case, the selection matrix $\boldsymbol{e}_{\mathbb{J}}$ comprises the final 7 columns of the 15-dimensional identity matrix (assuming the eight LISCC banks are ordered 1–8 in the model and the regional banks are ordered 9–15).

conditioning on LISCC banks' shocks or other regional banks' shocks.[17] We can then define aggregate spillover indices over these subsets:

$$
\begin{aligned}
jSOI_{L|L} &= \frac{1}{k_L}\sum_{i\in L} \mathcal{S}^{jnt}_{(\mathbb{J}=L)\to i}\,; \qquad jSOI_{L|R} = \frac{1}{k_L}\sum_{i\in L} \mathcal{S}^{jnt}_{(\mathbb{J}=R)\to i}\,;\\
jSOI_{R|L} &= \frac{1}{k_R}\sum_{i\in R} \mathcal{S}^{jnt}_{(\mathbb{J}=L)\to i}\,; \qquad jSOI_{R|R} = \frac{1}{k_R}\sum_{i\in R} \mathcal{S}^{jnt}_{(\mathbb{J}=R)\to i}\,;\\
jSOI_{\bullet|L} &= \frac{1}{K}\sum_{i=1}^{K} \mathcal{S}^{jnt}_{(\mathbb{J}=L)\to i}\,; \qquad jSOI_{\bullet|R} = \frac{1}{K}\sum_{i=1}^{K} \mathcal{S}^{jnt}_{(\mathbb{J}=R)\to i}\,;
\end{aligned}
\tag{15}
$$

where $\mathbb{J}=L$ denotes the set of LISCC banks and $\mathbb{J}=R$ denotes the set of regional banks. It should be understood that if bank i is a LISCC bank, then i is excluded from the conditioning set $\mathbb{J}=L$; if it is a regional bank, it is excluded from $\mathbb{J}=R$. $jSOI_{L|R}$ thus measures the average spillover to LISCC banks from the regional banks, and $jSOI_{R|L}$ is the average spillover to regional banks from the LISCC banks. $jSOI_{L|L}$ and $jSOI_{R|R}$ measure the mean spillovers *within* the LISCC banks and *within* the surviving regional banks, respectively. Lastly, $jSOI_{\bullet|L}$ is the mean spillover to all 15 banks in the sample from the LISCC banks, and $jSOI_{\bullet|R}$ is the mean spillover to all banks from the regional banks.

The non-diagonal entries down column j of $\boldsymbol{\zeta}^{gen}$ defined by Eq. (8) account for the bilateral transmission of shocks *from* variable j *to* others. There is no analogue in this case to the joint conditioning set of $\mathcal{S}^{jnt}_{\mathbb{J}\to i}$ in Eq. (12) because column j of $\boldsymbol{\zeta}^{gen}$ measures how the shocks of one variable — variable j — can individually explain the other variables' forecast error variances, whereas $\mathcal{S}^{jnt}_{\mathbb{J}\to i}$ measures how the shocks of several variables can explain the forecast error variance of variable i. However, it is informative to construct a metric that aggregates the total spillover from j to all others that is compatible with and comparable to Eqs. (12) and (13). To that end, we define the joint total spillover from variable j to all others as

$$
\mathcal{S}^{jnt}_{\bullet\leftarrow j} = \lambda \sum_{\substack{i=1\\ i\neq j}}^{K} \zeta^{gen}_{ij}, \tag{16}
$$

where

$$
\lambda = jSOI_{\bullet|\bullet} \Bigg/ \left(\frac{1}{K}\sum_{i=1}^{K}\sum_{\substack{j=1\\ j\neq i}}^{K} \zeta^{gen}_{ij} \right). \tag{17}
$$

Here, $\mathcal{S}^{jnt}_{\bullet\leftarrow j}$ is the diagonal-excluding sum of column j of $\boldsymbol{\zeta}^{gen}$ weighted by the scaling factor λ, and where λ is the ratio of the system-wide $jSOI_{\bullet|\bullet}$ to the average diagonal-excluding column sum (or row sum) of $\boldsymbol{\zeta}^{gen}$.[18] The scaling factor ensures that the VAR system is self-contained in the sense that the model's global measure of connectedness is the same whether we average over total spillovers *from* all others or total spillovers *to* all others:

$$
\frac{1}{K}\sum_{i=1}^{K} \mathcal{S}^{jnt}_{\bullet\to i} \equiv jSOI \equiv \frac{1}{K}\sum_{j=1}^{K} \mathcal{S}^{jnt}_{\bullet\leftarrow j}. \tag{18}
$$

Importantly, λ is a scalar applied to the entire $\boldsymbol{\zeta}^{gen}$ matrix, and is thus constant across all variables. As discussed in Caloia et al. (2019), a scalar normalization preserves the inherent asymmetries in the model's estimated parameters, unlike non-scalar normalizations that distort such asymmetries. Although it would be ideal to not require *any* normalizations in *any* metric, no such artificial asymmetries are introduced with our scalar normalization approach.[19]

Once we have both total "from others to i" and total "to others from i" spillovers, we compute the joint *net* total spillover for variable i as

$$
\mathcal{S}^{jnt}_{net,i} = \mathcal{S}^{jnt}_{\bullet\leftarrow i} - \mathcal{S}^{jnt}_{\bullet\to i}. \tag{19}
$$

$\mathcal{S}^{jnt}_{net,i} > 0$ implies that bank i is a net transmitter of spillovers, while $\mathcal{S}^{jnt}_{net,i} < 0$ implies that bank i is a net receiver of spillovers. Another way to conceptualize the global adding up constraint in Eq. (18) is that the joint net total spillovers, $\mathcal{S}^{jnt}_{net,i}$, must sum to zero.[20]

3. Estimation methods and results

Our primary objective is to estimate the effect of the bank failures and panic in March 2023 on the connectedness of surviving banks, using the spillover metrics described above. We treat the failures themselves — in particular, their timing — as an exogenous shock to the system as in an event study.

We employ two empirical approaches to compare spillovers before and after the failures. In the first, we estimate VAR models and compute the implied spillover metrics separately over two sub-periods of the data, where the sample break occurs when the March 2023 banking turmoil began. For the baseline results, the pre-failure period ranges from October 27, 2021 to March 6, 2023 and the post-failure period ranges from March 7, 2023 to July 12, 2024; a full sample of 680 observations is evenly split between the two periods. Given the extended time periods of the split samples, comparing the spillovers in the pre-failure and post-failure periods allows us to observe the *long-term* effects of the regional banking failures. The split sample spillovers are reported and discussed in Section 3.1.

The choice to split the sample on March 6–7, 2023 is based on our assessment of the timing of information flows regarding regional bank troubles. On March 8, two newsworthy announcements were made publicly that provided relevant information to stock markets. Silvergate Bank, a major player in cryptocurrency markets, announced its intent to voluntarily liquidate its assets and shut down operations (Edelman Smithfield, 2023). On the same day, Silicon Valley announced large losses on security sales and plans to raise new capital (Metrick, 2024).[21] News of Silicon Valley Bank's financial instability induced a bank run on deposits. Silicon Valley failed two days later, followed by Signature Bank after another two days. Because news of regional banking instability spread rapidly and markets are forward looking, we chose March 7 as the beginning of the post-failure period to account for the possibility that news began to hit markets a day prior to these official announcements.[22] This choice implies that the "post-failure" period includes the dates when the banks actually failed.

[17] Other work attempts to measure spillovers across broad subsets of the variables in the system. For example, Alter and Beyer (2014, section 3.4) estimate credit default swap price spillovers across banks and sovereign governments; Demirer et al. (2018, section 5.3.2) estimate separate indices for cross-country and within country connectedness of bank return volatility; and Greenwood-Nimmo et al. (2021) compute partially disaggregated spillover metrics using multiple macroeconomic variables for 25 countries. However, these studies do not entail the advantages of our joint conditioning set approach.

[18] Note that if $\lambda = \frac{1}{K-1}$, the spillover metric in Eq. (16) is the simple average of the off-diagonal elements of column j of $\boldsymbol{\zeta}^{gen}$.

[19] Indeed, if $\zeta^{gen}_{ij} > \zeta^{gen}_{ji}$, then $\lambda\zeta^{gen}_{ij} > \lambda\zeta^{gen}_{ji}$ will remain true as long as the scalar $\lambda > 0$. However, as explored in Caloia et al. (2019), the model's intrinsic asymmetries are distorted using the normalization of Diebold and Yilmaz (2012, 2014) where ζ^{gen}_{ij} and ζ^{gen}_{ji} are normalized by different row-sum values according to Eq. (9).

[20] Although not utilized directly in this paper, pairwise net spillover for variable i with respect to j could be computed as $\mathcal{S}_{net,i\leftrightarrow j} = \zeta^{gen}_{ij} - \zeta^{gen}_{ji}$.

[21] Silvergate, unlike Silicon Valley, did not formally fail and was not placed under the receivership of the FDIC, and its underlying problems were related to cryptocurrency markets and not interest rate volatility. However, Silvergate's voluntary liquidation announcement likely contributed to the general uncertainty in the banking system during this period. Adding to this uncertainty, albeit at a later date, was the collapse of Credit Suisse: on March 19, 2023, the bank announced it would be acquired by UBS Group with facilitation and intervention by Swiss regulators, thus marking the collapse of a globally systemically important bank and foreign exchange counterparty to the Federal Reserve.

[22] We examine robustness to this specification in Section 3.3.

The second strategy, standard in the connectedness literature, estimates the VAR model over rolling windows from January 2, 2019 to July 12, 2024. The rolling-window estimation strategy is consistent with a time-varying parameter stochastic process for which the parameters drift sufficiently slowly over time.[23] Our rolling estimation window is 250 days, approximately the number of trading days in a calendar year. The relatively long window length is a deliberately conservative choice that reduces noise in the spillover plots and allows us to better identify major changes to the spillovers across time. Importantly, careful inspection of the spillover results as the regional banking turmoil rolls into the window allows us to observe the *short-term* effect of the bank failures. Since our data sample begins on January 2, 2019, the 250-day window length implies that the first rolling window ends on December 27, 2019. This sample period allows us to compare the effects of the COVID-19 pandemic on banking spillovers to the effects of the bank failures three years later. Also note that this strategy does not require choosing a specific date to split the sample. These results are reported in Section 3.2.

Both strategies entail estimating Eq. (5) to obtain joint spillover metrics. For any given sample of banks and time period, we do so in the conventional way by estimating and inverting the VAR of the return volatilities over various time series samples. These estimates are sufficient for constructing all spillover metrics. The baseline model is a 15-variable VAR including the volatilities of the eight LISCC banks and the seven surviving regional banks. In all models, the VAR includes a constant term and one common lag across the system selected using BIC. When the spillover metrics are computed, we select the time horizon to be $H = 20$ days.

3.1. Split-sample average spillovers

Table 2 reports the joint spillover metrics (in percent) broken down by sample period and banking group. The first three columns show the total spillover from all other banks to the row bank following Eqs. (12), (13), and (14), where the conditioning set $\mathbb{J}$ includes shocks of all 14 other banks (excluding own shocks). The first column includes full sample results, from October 2021 to July 2024, while the second and third are results for the split samples.

According to the final row of the first column, the system's overall $jSOI_{\bullet|\bullet}$ for the entire sample period is estimated to be just over 75% — on average, shocks from other banks jointly explain more than three quarters of the 20-day forecast error variance of return volatility for the individual banks. This finding implies a large degree of connectedness in the banking system for this period, and is generally in line with other studies.[24] The average total spillover from all others to the LISCC banks in the top panel (row nine) is 74.2%, while the average total spillover from all others to the regional banks in the bottom panel (penultimate row) is slightly larger at 77.2%. The only meaningful difference between the banking groups over the full sample is that the range of the LISCC banks is smaller than for the regional banks : 70.7% to 81.4% compared with 67.5% to 85.3%.

What is the long-term impact of bank failures on system-wide spillovers? The overall $jSOI_{\bullet|\bullet}$ is only about 2.6 percentage points lower in the post-failure period than the pre-failure period. However, this relatively small average effect obscures the differential responses of systemically important and regional banks. Joint spillovers from all others *fall* by about ten percentage points on average for LISCC banks (from 78.3% to 68.2%) but *rise* by about six percentage points for regional banks (from 75.4% to 81.5%). The only LISCC bank not to exhibit a decline in spillovers from pre- to post-failure is Bank of New York Mellon, for which there is essentially no change. The only regional banks not to exhibit a rise are US Bank and Western Alliance, also essentially unchanged. On average and over the long run, LISCC banks have become less connected to the overall banking system since the 2023 banking panic, while regional banks have become more connected.

Now turn to the spillovers from conditioning on the different banking subsets of $\mathbb{J}$, as in Eqs. (12) and (15). The results are contained in the final six columns of Table 2. Over the full time period, the strongest connections are within groups. Shocks from LISCC banks explain 69.3% of the LISCC banks' volatility forecast error variance on average, while regional shocks explain only 42.5% of the LISCC bank's forecast error variance. On average, shocks from regional banks explain 73.6% of regional bank forecast error variance, whereas LISCC shocks explain only 42.6% of regional bank forecast error variance.[25] This pattern is generally uniform over individual banks.

Splitting the sample around the failure dates refines our inference about the effects of bank failures on spillovers. LISCC banks became more disconnected *regardless of the conditioning set*. The spillovers from all other LISCC banks to each individual LISCC bank on average fell from 71.1% to 58.2% (about 13 percentage points) after the panic. The spillovers from regional banks to LISCC banks fell from 59.9% to 40.8% (over 19 percentage points) after the panic. For regional banks, the mean spillover received from LISCC banks falls significantly after the panic from 59.7% to 46.8%, but spillovers to regional banks *rose* with respect to their own conditioning set from 65.6% to 70.3%. There is a *consolidation* of regional banks after the failures — the rise in connectedness among regional banks is not with respect to the system as a whole, but only with the set of other regional banks.

Fig. 3 illustrates these differences in a bar chart. Each purple (or brown) bar shows the difference before and after the failures for spillovers *coming from* the LISCC banks (or from the regional banks), excluding own shocks. As shown in the purple bars, spillovers from LISCC banks declined for *all* banks — both LISCC and regional — following the failures. As shown in the brown bars, spillovers from regional banks to all of the LISCC banks also fell. In contrast, spillovers from the regional banks to five of the regional banks — Comerica, Cullen/Frost, KeyBank, Synovus, and US Bank — increased in the aftermath of the crisis. Only Western Alliance and Zions Bank do not quite fit the pattern since their spillovers from other regional banks fell slightly after the failures. But as we will see in Section 3.2, Western Alliance and Zions Bank were major *transmitters* of volatility shocks to the other banks.

3.2. Time-varying spillovers

The split-sample results above allow for a one-time shift in the data generating process due to the 2023 bank failures and panic. Here, we allow for gradual drift in the parameters of the process to learn about spillover dynamics in the short run and their persistence in the longer run.

Fig. 4 reports the main results from the rolling window VAR models in three panels. The top panels plot estimates of the system-wide $jSOI_{\bullet|\bullet}$ in black and the indices from all banks to each banking group: LISCC banks ($jSOI_{L|\bullet}$ in purple) and regional banks ($jSOI_{R|\bullet}$ in brown). These are the comprehensive joint spillover indices of Eqs. (13) and (14). Each point on the time axis represents the end date of each 250-day rolling estimation window. The right-hand side panels zoom in on the failure dates for clarity.

The COVID-19 pandemic shock of March 2020 caused a rapid and remarkable rise in system-wide connectedness: all three of $jSOI_{L|\bullet}$, $jSOI_{R|\bullet}$, and $jSOI_{\bullet|\bullet}$, rose by roughly 20 percentage points from a starting point of just over 70% in mid-February 2020 to about 90% by

[23] For specific conditions, see Petrova (2019).

[24] For example, McNelis and Yetman (2020) estimate the $gSOI$ for a sample of globally important banks from 2007 to 2018 to be 78%.

[25] Owing to the cross-correlation of the shocks, these explanatory shares do not sum to 100%.

Table 2
Split sample joint spillover metrics.

	Full	Pre	Post	Full	Pre	Post	Full	Pre	Post
		$\mathcal{S}^{jnt}_{\bullet\to i}$			$\mathcal{S}^{jnt}_{(\mathbb{J}=L)\to i}$			$\mathcal{S}^{jnt}_{(\mathbb{J}=R)\to i}$	
BAC	81.41	86.51	72.99	77.32	79.76	65.18	51.55	65.30	47.80
BK	78.10	77.12	77.29	74.46	70.64	68.88	42.60	61.43	39.80
C	73.73	76.11	65.51	69.18	70.98	55.59	44.56	57.27	42.79
GS	71.75	73.58	65.44	66.90	66.73	60.41	36.75	50.03	32.63
JPM	70.71	80.52	58.46	63.54	73.03	48.12	35.59	62.42	32.23
MS	73.99	76.33	70.40	68.60	70.77	62.82	40.98	57.48	39.40
STT	71.60	78.16	65.02	67.91	68.15	53.60	41.28	64.11	39.96
WFC	72.18	77.88	70.24	66.38	68.87	51.29	47.03	61.29	51.64
LISCC		$jSOI_{L\|\bullet}$			$jSOI_{L\|L}$			$jSOI_{L\|R}$	
Mean	74.18	78.28	68.17	69.29	71.12	58.24	42.54	59.92	40.78
		$\mathcal{S}^{jnt}_{\bullet\to i}$			$\mathcal{S}^{jnt}_{(\mathbb{J}=L)\to i}$			$\mathcal{S}^{jnt}_{(\mathbb{J}=R)\to i}$	
CMA	85.31	78.45	90.04	41.43	61.06	44.51	83.77	70.73	80.26
CFR	67.54	64.15	73.98	41.03	46.22	41.12	64.33	56.10	66.73
KEY	84.38	78.15	88.56	41.51	57.95	49.78	81.82	72.56	78.14
SNV	83.72	79.17	86.37	52.93	62.00	50.71	80.83	71.84	77.79
USB	74.67	78.70	78.08	54.90	74.83	56.16	66.06	53.22	61.94
WAL	67.67	72.68	72.59	30.67	55.01	40.94	64.13	62.65	58.74
ZION	77.05	76.63	80.76	35.67	60.81	44.12	74.28	71.80	68.68
Regional		$jSOI_{R\|\bullet}$			$jSOI_{R\|L}$			$jSOI_{R\|R}$	
Mean	77.19	75.42	81.48	42.59	59.70	46.76	73.60	65.56	70.33
Overall		$jSOI_{\bullet\|\bullet}$			$jSOI_{\bullet\|L}$			$jSOI_{\bullet\|R}$	
Mean	75.59	76.94	74.38	56.83	65.79	52.88	57.04	62.55	54.57

The full sample period is October 27, 2021 to July 12, 2024. The pre-failure period is October 27, 2021 to March 6, 2023. And the post-failure period is March 7, 2023 to July 12, 2024. Spillover metrics are computed using Eqs. (12), (13), (14), and (15).

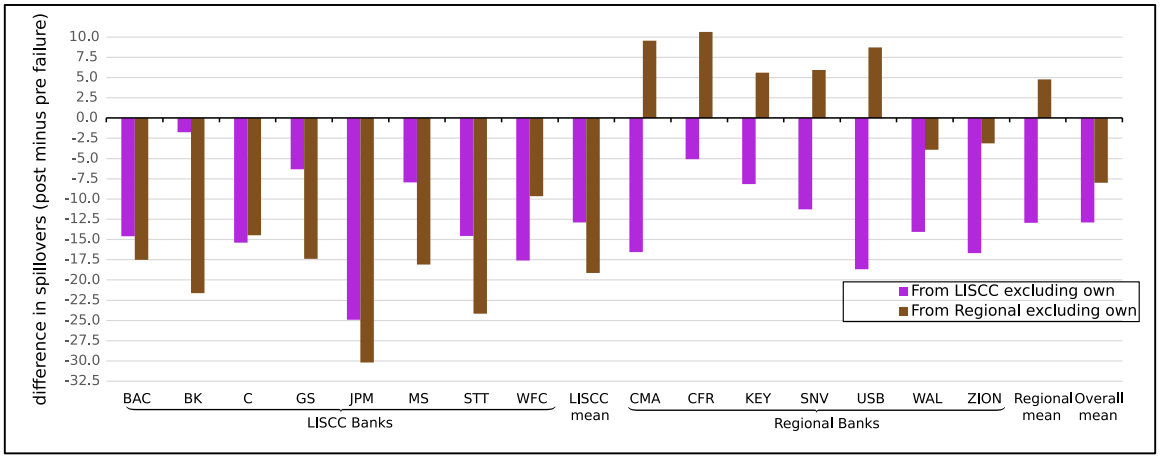


Fig. 3. Purple: $\mathcal{S}^{jnt}_{(\mathbb{J}=L)\to i}$ in the post-failure period minus $\mathcal{S}^{jnt}_{(\mathbb{J}=L)\to i}$ in the pre-failure period for each bank. Differences (post- minus pre-failure) for the LISCC mean ($jSOI_{L|L}$), the regional mean ($jSOI_{R|L}$), and the overall mean ($jSOI_{\bullet|L}$) are also shown. **Brown:** $\mathcal{S}^{jnt}_{(\mathbb{J}=R)\to i}$ in the post-failure period minus $\mathcal{S}^{jnt}_{(\mathbb{J}=R)\to i}$ in the pre-failure period for each bank. Differences (post- minus pre-failure) for the LISCC mean ($jSOI_{L|R}$), the regional mean ($jSOI_{R|R}$), and the overall mean ($jSOI_{\bullet|R}$) are also shown.

mid-March. They then exceed 90% for a year, until the initial periods of the pandemic roll out of the sample in March 2021. The uniformity of the COVID-19 response is also striking given that there is essentially no deviation of LISCC banks' $jSOI_{L|\bullet}$ and regional banks' $jSOI_{R|\bullet}$ from the system-wide $jSOI_{\bullet|\bullet}$ during the pandemic period. During the period after COVID-19, $jSOI_{L|\bullet}$, $jSOI_{R|\bullet}$, and $jSOI_{\bullet|\bullet}$ all gradually hover around 80% on average — a level about ten percentage points higher than pre-pandemic levels — and the relatively small spread between LISCC and regional banks that existed prior to the pandemic reappears.

As noted above, these findings are generally consistent with those of Tabak et al. (2022, Figure 7) for their global sample. However, we find a more persistent rise among US banks than that earlier result, even though the estimation windows are similar.[26] That paper's choice to focus on national banking aggregates, while important for the questions it asks, is not able to uncover the effects of COVID-19 we find for connectedness among different types of banks in the US.

The paths of $jSOI_{L|\bullet}$, $jSOI_{R|\bullet}$, and $jSOI_{\bullet|\bullet}$ are altered by the banking panic of 2023. Each of the three indices in the top panel exhibits a noticeable and abrupt increase immediately after the March 10–12, 2023 bank failures, indicative of significant movements in all bank returns in the face of Silicon Valley and Signature's collapse. However, the magnitude of the sudden change in spillovers was not uniform across the banking groups. From March 8 to 14, 2023, the LISCC banks' $jSOI_{L|\bullet}$ rose by 5.6 percentage points, but the regional banks' $jSOI_{R|\bullet}$ rose by 12.7 percentage points. We also observe a small jump of 2.5 percentage points in the total spillovers to the regional banks following First Republic's failure in May; but no such changes are observed with the LISCC banks in May. Furthermore, while total spillovers to LISCC banks gradually but significantly declined in the 12 months after the 2023 failures, the total spillovers to the regional banks remain elevated. Toward the end of the overall estimation period

[26] Their window is 130 observations with a two-day data frequency; thus, both studies use roughly one-year estimation windows.

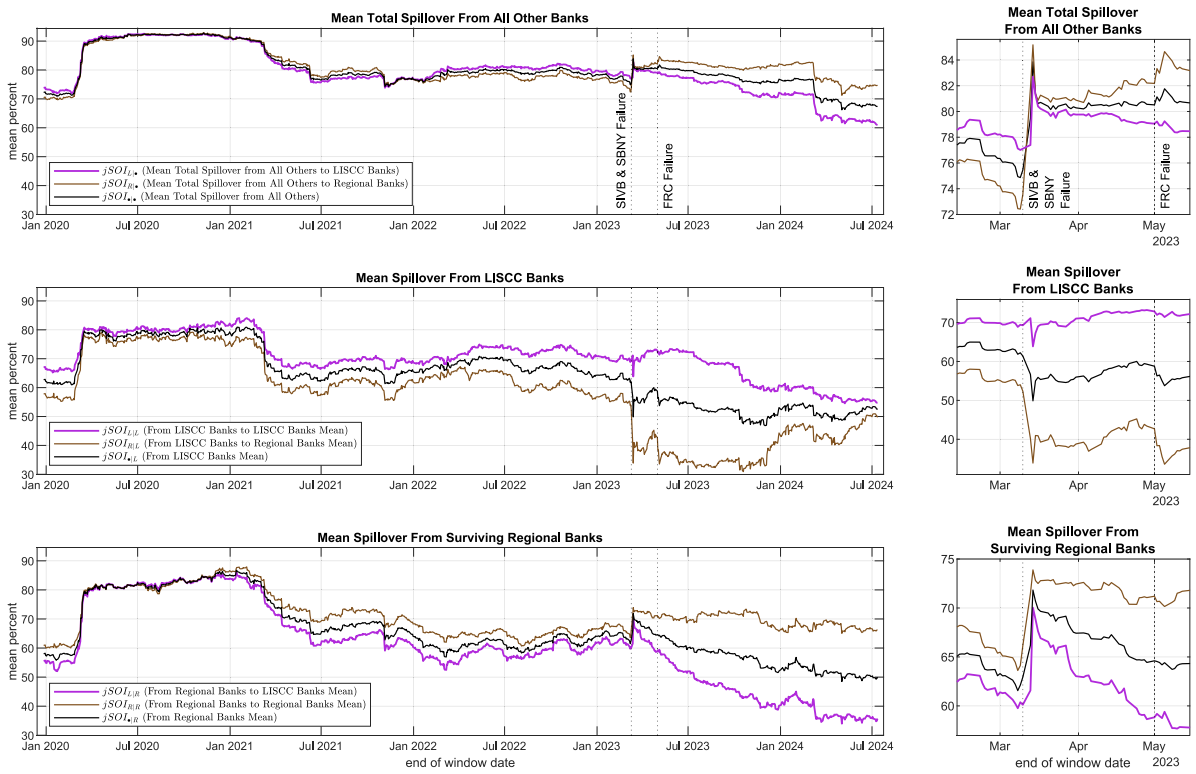


Fig. 4. Joint spillover metrics over rolling windows. The left-hand panels span the full time period. The right-hand panels zoom in on the spring 2023 crisis. Note the different scales used in the right-hand panels. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

in July 2024, the gap between regional and LISCC bank connectedness is almost 14 percentage points, far exceeding that of any prior period in our sample (and especially during the pandemic episode). These results are generally consistent with those of our split-sample strategy, although they suggest that connectedness largely spiked after Silicon Valley's failure, and not on March 8 with the announcements of Silvergate's liquidation and Silicon Valley's bond portfolio loss.

The middle panels of Fig. 4 report spillovers from LISCC banks' shocks ($jSOI_{L|L}$, $jSOI_{R|L}$, and $jSOI_{\bullet|L}$). Immediately following the March failures, spillovers from LISCC banks decline for both banking groups, although the decline is extremely temporary (just a few days) for LISCC banks but persistent and vastly larger for regional banks. Between March 8 and 14, 2023, the $jSOI_{R|L}$ fell from 53.8% to 33.9%. Those from LISCC to regional spillovers rose briefly in April 2023 before falling again in early May when First Republic failed. From May to November 2023, the $jSOI_{R|L}$ remained below 38% — drastically lower than the 60% level it hovered around throughout 2022. For most of 2023 post-failures, the difference between the "from LISCC to regional" and "from LISCC to LISCC" spillovers was large. However, by late 2023 this difference begins to diminish as regional bank spillovers from LISCC shocks rise and LISCC to LISCC bank spillovers trend downward. By the end of the sample in July 2024, there is only about a five percentage point gap between the purple $jSOI_{L|L}$ and brown $jSOI_{L|R}$ lines, which is even smaller than it was throughout almost all of 2021 and 2022. So while the link from LISCC bank shocks to regional banks fell significantly post-failure, the decline relative to LISCC banks' connections within themselves appears to be only a medium-term transitory phenomenon.

This is not the case for the spillover indices with respect to regional bank shocks — $jSOI_{L|R}$, $jSOI_{R|R}$, and $jSOI_{\bullet|R}$ in the bottom panels of Fig. 4. Spillovers from regional to regional banks rise by about 10.3 percentage points following the March failures and remain consistently elevated. In contrast, spillovers from regional to LISCC banks still rise in March but then gradually fall over the post-failure period. When the bank failure-induced spillovers peaked on March 14, 2023, the $jSOI_{R|R}$ was 3.9 percentage points above the $jSOI_{L|R}$; by First Republic's failure on May 1, the gap widened to 12.4 percentage points. And by the end of the sample period in July 2024, the gap between $jSOI_{R|R}$ and $jSOI_{L|R}$ was 31 percentage points — much larger than at any point in the pre-failure period.

The panels of Fig. 4 also illustrate the effects of the May 1 failure of First Republic. In terms of spillover indices, this bank's failure is *mostly* a non-event. The only apparent results are a decline in spillovers from LISCC to regional banks ($jSOI_{R|L}$) when First Republic failed and a small increase in total spillovers from all other banks to regional banks ($jSOI_{R|\bullet}$). But otherwise, there is very little effect. The clear implication is that First Republic's failure was largely anticipated, conditional on the troubles of Silicon Valley and Signature banks.

Whereas Fig. 4 focuses on regional and LISCC bank groupings, Fig. 5 examines the time-varying spillovers for each individual bank. In each panel, the red curve is the joint total spillover from all other banks to bank i ($\mathcal{S}^{jnt}_{\bullet\rightarrow i}$ from Eq. (12) when $\mathbb{J}$ consists of all banks' shocks excluding own), the black curve is the joint total spillover from bank i to all others ($\mathcal{S}^{jnt}_{\bullet\leftarrow i}$ in Eq. (16)), while the shaded pink and gray areas are *net* total spillovers ($\mathcal{S}^{jnt}_{net,i}$ in Eq. (19)): gray means that bank i is a net transmitter of spillovers ($\mathcal{S}^{jnt}_{net,i} > 0$); pink means that bank i is a net receiver ($\mathcal{S}^{jnt}_{net,i} < 0$). The vertical dotted lines indicate failure dates.

During the 2020 COVID-19 period, the uniformity of the increase in spillovers from all others across banking types is evident from the red curves in each panel. $\mathcal{S}^{jnt}_{\bullet\rightarrow i}$ for all banks showed a marked rise in February–March 2020 and remained elevated for about a year. However, there is substantial variation in total spillovers *to all others* from the individual banks ($\mathcal{S}^{jnt}_{\bullet\leftarrow i}$), and thus variation in net total spillovers ($\mathcal{S}^{jnt}_{net,i}$). During the roughly one-year COVID period, five of the eight LISCC banks were clear net receivers of volatility spillovers (Bank of New York Mellon, Citigroup, Goldman Sachs, Morgan Stanley, and State Street), while two were clear net transmitters (JP Morgan Chase and Wells Fargo).[27] Of the regional banks, Cullen/Frost, KeyBank,

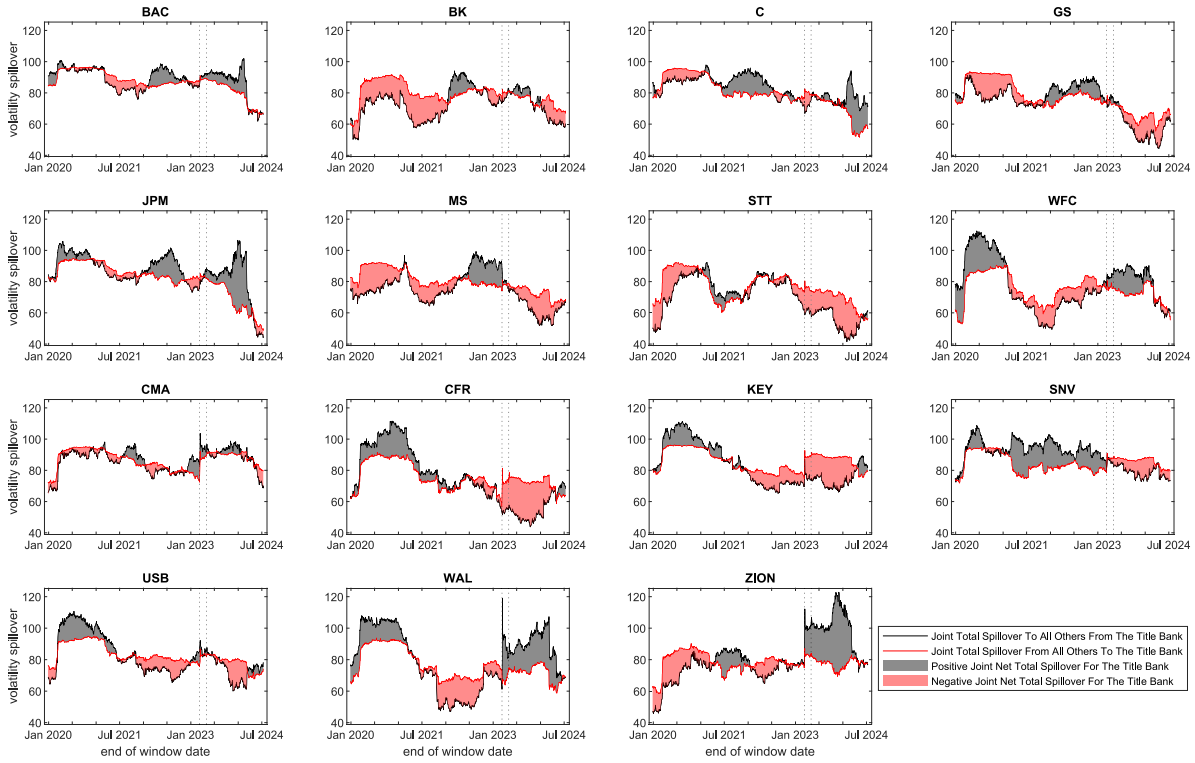


Fig. 5. Joint total spillover to all others from the title bank ($\mathcal{S}^{jnt}_{\bullet\leftarrow i}$), joint total spillover from all others to the title bank ($\mathcal{S}^{jnt}_{\bullet\rightarrow i}$), and joint net total spillovers ($\mathcal{S}^{jnt}_{net,i}$) estimated over 250-day rolling windows. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Synovus, US Bank, and Western Alliance were all net transmitters of volatility spillovers, while only Comerica and Zions Bank were net receivers.

With respect to the banking panic of 2023, each of the LISCC banks experiences a gradual decline in its joint spillovers *from all others* in the months following the failures. The index for JP Morgan Chase depreciates the most mid-2023 to mid-2024. However, the *immediate* impact of the March 2023 failures is barely noticeable for all LISCC banks.[28]

For regional banks, the effects of the failures are very similar for Comerica, Cullen/Frost, KeyBank, Synovus, and US Bank: a sharp immediate increase in the red $\mathcal{S}^{jnt}_{\bullet\rightarrow i}$ following the failures and then a steady and persistently elevated level until the crisis events roll out of the sample. This pattern is less obvious for Western Alliance and Zions Bank. These findings underlie the results of Fig. 4 but show that there is greater dispersion among regional banks than among LISCC banks.

By focusing on the black $\mathcal{S}^{jnt}_{\bullet\leftarrow i}$ lines, we can see who the primary transmitters of volatility shocks are in the immediate aftermath of the March 2023 failures. Again, the dispersion across regional banks is noteworthy. Between March 8–14, 2023 total spillovers coming from Comerica rose 20.1 percentage points, total spillovers from Zions bank rose 35.6 percentage points, and spillovers from Western Alliance rose the most at 55.6 percentage points. As indicated by the gray shaded regions, Western Alliance and Zions Bank remained net transmitters of spillovers in the year following the failures. The fact that Western Alliance and Zions Bank sent extremely large shocks to the other banks (especially the other regional banks) helps explain why those two banks did not follow the same pattern as the other regional banks in Fig. 3.

3.3. Robustness

Our main findings are generally robust to our model specifications. None of our findings change significantly when we estimate VAR models with two or three lags, rather than one, or compute spillovers for horizons $H = 10$ or $H = 30$, rather than the baseline of $H = 20$. The results are qualitatively unaltered by using rolling regression window lengths of 150, 200, and 300 days, rather than the baseline of 250 days.[29] These are standard checks.

We also considered robustness to other assumptions unique to our exercise. First, we examined sensitivity to the dates at which we split the sample in Section 3.1. To do so, we hold fixed the 680-day full sample period, but alter the break point from the baseline case of March 6–7, 2023. We try February 28, March 8, and March 10 as alternate dates to end the pre-failure subperiod, and where the following trading day begins the post-failure subperiod. Note that the last case puts Silicon Valley's failure on March 10 into the first (pre-failure) subperiod, so this check is an important one to ensure that single shock is not driving the entire results. Table 3 — which shows the split-sample spillovers using March 10–13 to divide the sample — illustrates that this is not the case, since the basic inference holds up. Agreeing with Table 2, Table 3 shows that mean spillovers from LISCC banks to other LISCC banks ($jSOI_{L|L}$), mean spillovers from LISCC banks to regional banks ($jSOI_{R|L}$), and mean spillovers from regional banks to LISCC banks ($jSOI_{L|R}$) all fell in the post-failure period. Critically, mean spillovers from regional banks to other regional banks ($jSOI_{R|R}$) increased in the post-failure period according to both Tables 2 and 3. The difference between regional banks' spillovers among themselves and all other categories of spillovers remains apparent.

[27] Bank of America's net spillovers were close to zero during the pandemic.
[28] Perhaps the one exception is a decline in spillovers to all others from Morgan Stanley in March 2023.
[29] In some cases, the path of the joint spillover metrics were partially dependent on the rolling window length because that length determines when the failure event rolled out of the sample. However, in no cases were the differences between LISCC and regional banks affected.

Table 3
Split-sample joint spillover metrics: Robustness to division dates.

	Full	Pre	Post	Full	Pre	Post	Full	Pre	Post
		$S^{jnt}_{\bullet\to i}$			$S^{jnt}_{(\mathbb{J}=L)\to i}$			$S^{jnt}_{(\mathbb{J}=R)\to i}$	
BAC	81.41	86.62	70.54	77.32	79.83	64.29	51.55	66.39	46.91
BK	78.10	76.74	74.89	74.46	70.46	67.67	42.60	61.02	38.66
C	73.73	76.40	62.56	69.18	71.33	53.49	44.56	57.30	40.73
GS	71.75	73.79	63.67	66.90	66.92	59.37	36.75	49.72	31.50
JPM	70.71	80.52	56.63	63.54	73.09	46.26	35.59	62.89	31.41
MS	73.99	75.84	68.09	68.60	70.60	61.78	40.98	56.39	38.24
STT	71.60	77.63	61.69	67.91	67.57	51.54	41.28	62.56	38.01
WFC	72.18	77.66	65.88	66.38	68.99	48.45	47.03	62.14	49.60
LISCC		$jSOI_{L\|\bullet}$			$jSOI_{L\|L}$			$jSOI_{L\|R}$	
Mean	74.18	78.15	65.49	69.29	71.10	56.61	42.54	59.80	39.38
		$S^{jnt}_{\bullet\to i}$			$S^{jnt}_{(\mathbb{J}=L)\to i}$			$S^{jnt}_{(\mathbb{J}=R)\to i}$	
CMA	85.31	79.30	86.12	41.43	61.35	40.34	83.77	72.39	80.11
CFR	67.54	65.11	68.97	41.03	45.90	38.12	64.33	57.73	63.75
KEY	84.38	78.14	85.51	41.51	58.31	46.99	81.82	72.85	78.48
SNV	83.72	80.03	84.28	52.93	61.99	47.34	80.83	73.32	79.62
USB	74.67	78.95	76.17	54.90	75.14	54.30	66.06	54.95	62.79
WAL	67.67	71.46	66.60	30.67	47.57	34.69	64.13	64.06	59.81
ZION	77.05	75.56	80.65	35.67	57.58	41.02	74.28	72.36	70.83
Regional		$jSOI_{R\|\bullet}$			$jSOI_{R\|L}$			$jSOI_{R\|R}$	
Mean	77.19	75.51	78.33	42.59	58.26	43.26	73.60	66.81	70.77
Overall		$jSOI_{\bullet\|\bullet}$			$jSOI_{\bullet\|L}$			$jSOI_{\bullet\|R}$	
Mean	75.59	76.92	71.48	56.83	65.11	50.38	57.04	63.07	54.03

The pre-failure period is October 27, 2021 to March 10, 2023. The post-failure period is March 13, 2023 to July 12, 2024. Spillover metrics are computed using Eqs. (12), (13), (14), and (15). Full sample results are unchanged from the main results in Table 2 but are maintained here for easy reference.

Table 4
Split-sample joint spillover metrics: Robustness to including data from failed banks.

	$S^{jnt}_{\bullet\to i}$	$S^{jnt}_{(\mathbb{J}=L)\to i}$	$S^{jnt}_{(\mathbb{J}=R)\to i}$
BAC	86.56	77.96	67.28
BK	78.10	67.93	65.02
C	77.75	68.53	61.75
GS	74.99	65.21	55.58
JPM	81.57	70.27	65.24
MS	78.04	69.52	60.83
STT	79.21	66.31	68.68
WFC	80.16	66.87	65.59
LISCC	$jSOI_{L\|\bullet}$	$jSOI_{L\|L}$	$jSOI_{L\|R}$
Mean	79.55	69.08	63.75
	$S^{jnt}_{\bullet\to i}$	$S^{jnt}_{(\mathbb{J}=L)\to i}$	$S^{jnt}_{(\mathbb{J}=R)\to i}$
CMA	80.24	62.96	73.20
CFR	65.28	46.93	56.88
KEY	78.75	60.00	73.14
SNV	79.99	62.72	73.62
USB	78.73	72.86	56.70
WAL	77.55	54.80	70.26
ZION	78.02	60.04	73.81
SIVB	64.53	33.09	56.86
SBNY	61.55	41.58	57.15
FRC	56.55	45.84	50.73
Regional	$jSOI_{R\|\bullet}$	$jSOI_{R\|L}$	$jSOI_{R\|R}$
Mean	72.12	54.08	64.24
Overall	$jSOI_{\bullet\|\bullet}$	$jSOI_{\bullet\|L}$	$jSOI_{\bullet\|R}$
Mean	75.42	60.75	64.02

Estimated using an 18-variable VAR with one lag over the pre-failure period of October 27, 2021 to March 6, 2023.

A more interesting check is whether our main findings depend on the assumption that failed banks do not belong in our pre-failure sample. That assumption has the advantage of focusing our research question on how the network of existing and ongoing banks responded to the "event" marked by the 2023 regional bank failures. However, this modeling choice excludes potentially informative volatility data from the failed banks, and could over- or under-state connectedness measures in the pre-failure period. For example, if system-wide spillovers are understated pre-failure, then estimated changes in spillovers post-failure would be upward biased (and vice versa).[30]

To examine this question, we estimate an expanded VAR model during the pre-failure period that includes the return volatility data for Silicon Valley, Signature, and First Republic banks along with the eight LISCC banks and seven surviving regional banks in the baseline sample. If our spillover comparisons before and after the failures from the expanded 18-variable system are sufficiently close to those of the baseline model, we can be confident in the robustness of our results and inference.

Table 4 reports the joint spillover metrics for the period October 27, 2021 to March 6, 2023 for the 18-bank expanded sample. Recall from Table 2 that average spillovers from LISCC banks to other LISCC banks fell from 71.1% to 58.2% pre- to post-failure. When the failed banks are included in the pre-failure sample, as in Table 4, the decline is from 69.1% to (the same) 58.2%. Average spillovers from regional banks to LISCC banks fall from 59.9% to 40.8% for the baseline sample, but 63.8% to 40.8% in the expanded sample. Average spillovers from LISCC banks to regional banks fall from 59.7% to 46.8% for the baseline sample, but 54.1% to 46.8% in the expanded sample. Finally, mean regional to regional bank spillovers still exhibit an increase from pre- to post-failure, but the change is slightly larger: the within regional banks spillovers increased from 65.6% to 70.3% in the baseline sample (a 4.7 percentage point change) and increased from 64.2% to 70.3% in the expanded sample (a 6.1 percentage point change).

Thus, while the magnitudes of the comparisons slightly change (on the order of 2 to 5 percentage points), the main finding — that only the subset of regional banks exhibits an increase in connectedness after the failures — is robust. Moreover, the pre-failure system-wide connectedness of $jSOI_{\bullet|\bullet} = 76.9\%$ in the baseline sample is only 1.5 percentage points different from that of the expanded sample's $jSOI_{\bullet|\bullet} = 75.4\%$. This proximity suggests that there is no large system-wide bias in the pre-failure results due to the exclusion of the failed banks.

[30] In most applications, the cross-sectional sample is well-defined and constant. However, in some cases, the cross-sectional sample is less clear and may even change over time. The issue of how to estimate spillovers as variables exit or enter the model is an important issue for future work using our methods.

4. Conclusion

In terms of nominal assets placed under receivership by the FDIC, the collapse of First Republic, Silicon Valley, and Signature banks in the spring of 2023 marked the second, third, and fourth largest bank failures in US history. The cause of each was a classic bank run on uninsured deposits. The last time the US banking system saw failures of this magnitude was the collapse of Washington Mutual during the 2008 financial crisis. In fact, the total nominal assets of First Republic, Silicon Valley, and Signature banks *exceeded* those of all banks that failed in 2008 (FDIC, 2024).

This paper uses the joint spillover metrics of Lastrapes and Wiesen (2021) and joint forecast error variance decomposition of Wiesen and Beaumont (2024) to examine the effects of these well-publicized regional bank failures on banking system connectedness. We look at daily stock return volatility of publicly traded banks to quantify the importance of shocks from other banks in explaining forecast error variance of individual banks and groups of banks in the VAR system. One of the main contributions of this paper is to use this approach to focus attention on the "often-ignored regional banking sector" (Balasubramanyan and Bianco, 2015).

We find that, on average over the period 2021 to 2024, US banks have been highly connected in the sense that most of the variation of their daily return volatility can be explained by shocks from other banks, on the order of 76%. This finding holds for a sample consisting of large, nation-wide banks deemed by regulators as systemically important (LISCC banks) and not-as-large regional banks. Before the 2023 failures, the behavior of LISCC banks' and regional banks' volatility spillovers were generally similar, especially during the COVID-19 pandemic of 2020. However, the post-failure paths of volatility spillovers for these two groups of banks have diverged substantially.

Using an estimation strategy that yields time-varying connectedness statistics, we find that the failures of Silicon Valley and Signature banks in mid-March 2023 led to a substantial increase in volatility spillovers to regional banks. However, this rise in spillovers to regional banks came only from other *regional* banks, whereas spillovers from LISCC banks fell significantly. Relative to a 12.7 percentage point rise in total spillovers received by regional banks in the immediate aftermath of the failures, LISCC banks received a much smaller rise in volatility transmissions — only 5.6 percentage points more. When First Republic failed in May 2023, its failure had a relatively minor impact and only affected other regional banks. As of the end of our sample period, the spillovers within regional banks remained elevated, whereas spillovers within LISCC banks, spillovers from LISCC to regional banks, and spillovers from regional to LISCC banks declined relative to pre-failure levels. These long-term effects are confirmed by both split-sample and rolling window results.

Questions regarding the magnitude of spillovers and contagion in the banking industry are historically and contemporaneously relevant, given the obvious implications for systemic risk in the financial system. Our finding that the systemically important LISCC banks were significantly less affected by the 2023 bank failures suggests that regulatory policy toward those banks — that they will be protected in periods of uncertainty — is credible. This conclusion is especially notable as some of the LISCC banks had uninsured deposit ratios similar to Silicon Valley and Signature banks.[31] The additional regulatory supervision of and restrictions placed on the LISCC banks (e.g., stress tests and capital requirements) appear to have had the intended effect of enhanced resilience — at least in this instance.

However, our finding that regional banks became more sensitive to volatility shocks after the 2023 failures, and that sensitivity appears long-lasting, suggests that regulators may need to shift attention to the stability of mid-sized financial institutions since they can pose significant risks to the broader financial system. Regulators might reconsider the scope and intensity of regulatory oversight for regional banks, including enhanced stress testing, stricter liquidity requirements, and targeted resolution planning tailored to their unique risk profiles and connections to other financial institutions. They might also consider a tiered regulatory approach, imposing requirements and deposit insurance premiums based upon the bank's connectedness in addition to the existing criteria of size, complexity, risk category, and CAMELS rating.[32] Better mechanisms for early detection of contagion risk and increased transparency about regional bank spillovers can build resilience in this increasingly significant component of the US financial system. As with all policies, this enhanced oversight should be weighed against the potential costs of additional regulatory burden, especially to the not-as-large banks under consideration. We leave specific policy recommendations as a critical topic for future research.[33]

CRediT authorship contribution statement

William D. Lastrapes: Writing – review & editing, Writing – original draft, Formal analysis, Conceptualization. **Thomas F.P. Wiesen:** Writing – review & editing, Writing – original draft, Formal analysis, Data curation, Conceptualization, Software, Methodology.

Data availability

Data will be made available on request.

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[31] Of the regional banks, Silicon Valley and Signature banks had uninsured deposit ratios of 93.8% and 89.3%, respectively. Of the systemically important LISCC banks, Bank of New York Mellon and State Street had uninsured deposit ratios of 92.0% and 91.2% respectively. See Table 1 and Hayes (2023).

[32] Recent work on deposit insurance premiums includes Oosthuizen and Zalla (2024), Kim and Rezende (2023), Camara et al. (2020), and Demirgüç-Kunt et al. (2015). Current insurance premiums are described in FDIC (2023).

[33] Such research would be especially timely in light of the recent report by Office of the Inspector General (2024) on the FDIC's readiness to resolve large regional banks that fail.

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